

TRACE-IDS — How to Explain My Project

Prep script for the recommendation meeting. Read it a few times so you can say it in your own words, without notes.

1. The one-line pitch (say this first)

"I tested whether a cyber-attack detector trained on one network works on a different network. It completely fails — and I found a cheap fix."

2. The 30-second version

"People train attack detectors on one dataset and report 95%+ accuracy. But a real system runs on a network it never saw. I trained on one real network dataset (NF-CICIDS2018) and tested on a different one (NF-UNSW-NB15). It caught zero attacks — it marked everything safe. The 95% accuracy was fake, because the new network is 96% safe traffic anyway."

"Then I found the fix: give the model just a few labeled examples from the new network. Five examples per class brought detection to 88%, ten to 93%, fifty to 99%. So you do not need to relabel everything — a tiny amount of data redeploys a working detector."

3. The honest finding (professors respect this part)

"I first tried the popular fixes — rescaling features, balancing classes, and using SHAP to pick transferable features. None worked; detection stayed at zero. I reported that honestly, then showed what actually works: a few labeled target examples."

4. Why I did it (if asked 'why this topic?')

"I am applying to the MS/PhD and wanted real research in your area — cross-dataset and cyber ML. I saw your work on network attack detection, so I built this to learn the methods and show I can work independently."

5. Simple answers to likely questions

If they ask...	You say...
What are the datasets?	NF-CICIDS2018-v3 and NF-UNSW-NB15-v3 — two standard NetFlow network-attack datasets that share the same 47 features.
What model?	A Random Forest. Simple on purpose — the point is the transfer problem, not a fancy model.
Main result?	Zero-shot: the detector caught 0% of attacks on the new network. Few-shot: 5 labeled examples per class recovered 88%, 50 gave 99%.
Why did accuracy look fine?	The new network is 96% safe traffic, so 'mark everything safe' scores 96% while catching nothing. That is why I report recall, not accuracy.
What does SHAP show?	Which features the model relies on — and that they differ between the two networks, which is why a source-only model does not transfer.

Limits?	Two datasets, a Random Forest, offline test. No real-time, no attacker-evasion. It is in the paper.
What is next?	A leave-one-dataset-out benchmark across more NetFlow datasets, and few-shot with deeper models.

6. If I freeze — the fallback sentence

“A detector trained on one network catches nothing on a new one, but five to ten labeled examples bring detection back to about 90%.”

Two tips

- Do not claim it is deployable or perfect — say ‘it is a study, with clear limits.’ Honesty beats hype.
- Open the paper’s references once so you can say ‘I read the related work.’